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SUPERSEDING
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MILITARY SPECIFICATION

TRACTORS, WHEELED, INDUSTRIAL, PNEUMATIC-TIRED, GASOLINE- AND DIESEL-ENGINE-DRIVEN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gasoline- and diesel-engine-driven, pneumatic-tired, 4-wheel-type, heavy-duty tractors and their attachments.

1.2 Classification. The tractors shall be of the following classes and sizes as specified (see 6.2 and 6.5):

Class 1 - Gasoline-engine-driven
Class 2 - Diesel-engine-driven

	Drawbar HP (without ballast)
Size R-10 (class 1 only)	5-10
Size R-15 (class 1 only)	11-15
Size R-20 (class 1 only)	16-20
Size R-25 (class 1 only)	21-25
Size R-30 (class 1 only)	26-30
Size R-35	31-35
Size R-40	36-40
Size R-45	41-45
Size R-50	46-50
Size R-55	51-55
Size R-60	56-60

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-P-455	- Plate, Photographic (Photosensitive Anodized Aluminum).
QQ-C-320	- Chromium Plating (Electrodeposited).
UU-T-81	- Tags, Shipping and Stock.
PPP-B-636	- Box, Fiberboard.

Military

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	- Plates; Identification, Transportation-Data and Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Heavy Duty.
MIL-T-3351	- Tractor, Full-Tracked, Low Speed; Tractor, Wheeled, Agriculture; Tractor, Wheeled, Industrial: and Their Attachments, Packaging of.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated (15-Ounce and 21-Ounce Capacity).
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-E-11275	- Engines, Gasoline, Industrial Type, General Specification for.
MIL-E-11276	- Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.

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MIL-I-11683	- Interference Suppression, Radio, Requirements for Engine Generators and Miscellaneous Engines.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-45199	- Lubricating Oil, Internal Combustion Engine (High Output Diesel).

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-448	- Test Methods for Construction and Industrial Machinery.
MS15001	- Fittings, Lubrication (Hydraulic) Surface Check, 1/4-28 Taper Threads, Steel, Type I.
MS15002	- Fittings, Lubrication (Hydraulic) Surface Check, Straight Threads, Steel, Type II.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

DRAWINGS

MO(ERDL)

D13200E7450	- Lamp Housing Resilient Mounting; for Construction Machinery.

MO(ATAC)

B209297	- Lamp, Trouble.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y. 10017.)

TIRE AND RIM ASSOCIATION, INCORPORATED

Tire and Rim Association Yearbook, with Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 34 North Hawkins Avenue, Akron, Ohio 44313.)

INDUSTRIAL PRESS

Machinery's Handbook.

(Application for copies should be addressed to the Industrial Press, 148 Lafayette Street, New York, N. Y.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D1692 - Methods of Test for Flammability of Plastics Foams and Sheeting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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OFFICIAL CLASSIFICATION COMMITTEE.

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The tractors shall be gasoline- or diesel-engine-driven, pneumatic-tired, and shall have two front steerable wheels and two rear drive wheels. The tractors shall be complete with all accessories specified herein and shall be for use with attachments such as front-end loaders, backhoes, mowers, and for drawbar work. All accessories supplied commercially as standard equipment shall be furnished. All tractors and attachments shall be the supplier's current standard production models except as specified herein.

3.2 Preproduction model. Unless otherwise specified (see 6.2), as soon as practicable after award of the contract, the supplier shall furnish a tractor to prove, prior to starting production, that his production methods and choice of design detail will produce tractors that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish tractors conforming to this specification.

3.3 Initial production tractors. When specified (see 6.2), an initial production tractor selected at random by the Government from the first 10 tractors being produced from production tooling shall be examined and tested as specified in 4.4 to determine conformance to the requirements of this specification. Acceptance of the initial production tractor shall not exclude the remaining tractors from inspection and acceptance levels specified in section 4.

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3.4 Material. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials selected by the supplier shall be compatible with the intended use in accordance with the engineering standards of the industrial and construction machinery industry and as set forth in such publications as ASTM standards, the SAE Handbook, and the Machinery's Handbook. Unless otherwise specified herein, rubber products furnished shall not exceed 5 calendar quarters in age from the date of cure of the rubber to date of acceptance of the tractor by the Government.

3.4.1 Threaded parts. Unless otherwise specified herein, threaded parts shall be in accordance with Handbook H-28.

3.5 Environmental and use conditions.

3.5.1 Operating temperature. Without preheating, the engine shall start within 5 minutes and the tractor shall operate under full load conditions within 15 minutes after engine start in any ambient temperature from minus 25° F. to plus 115° F. At low temperature, all starting and operating functions shall be performed by operators wearing arctic mittens and boots.

3.5.2 Use. The tractors and specified attachments shall withstand the usage encountered in normal industrial operations such as backhoe, loader, and mower operations without malfunction, damage, or permanent deformation. The tractors shall negotiate 30 percent side slopes and 30 percent longitudinal slopes or any combination thereof without any wheel of the tractor losing contact with the ground; and the engine shall maintain constant operating oil pressure and operate without malfunction under these slope conditions when tested as specified in 4.7.5.

3.6 Safety. All exposed parts which are subject to high operating temperature or which are energized electrically shall be insulated, enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions.

3.7 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair and replacement. Covers or plates which must be removed for component adjustment, repair,

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replacement, or maintenance shall be equipped with quick-disconnect fastenings where practicable. Drain outlets shall be located for accessibility and shall permit the use of receptacles for collecting drainage.

3.8 Engine.

3.8.1 Gasoline engine. Unless otherwise specified herein, the size and type of gasoline engine shall be as recommended by the supplier. When specified (see 6.2), the engine shall conform to MIL-E-11275, group II, liquid cooled. All engines shall conform to the applicable provisions specified herein.

3.8.1.1 Ignition system. Unless a Qualified Products List (QPL) engine is furnished, the ignition system shall be the manufacturer's standard.

3.8.2 Diesel engine. Unless otherwise specified herein, the size and type of diesel engine shall be as recommended by the supplier. When specified (see 6.2), the engine shall conform to MIL-E-11276. All engines shall conform to the applicable provisions herein.

3.8.3 Power and speed ratings. Engines conforming to MIL-E-11275 and MIL-E-11276 shall not require horsepower in excess of the intermittent brake horsepower rating of the engine for the applicable speed as established by the Qualified Products Lists when operated as specified herein. Commercial engines shall not exceed 90 percent of the maximum brake horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer when operated as specified herein.

3.8.4 Fuel system. Unless a QPL engine is furnished, the fuel system shall be the supplier's standard. The fuel system shall be equipped with a removable fuel sediment trap and shutoff valve.

3.8.5 Cranking system. Unless a QPL engine is furnished, the electric power cranking system shall be the supplier's standard. When specified (see 6.2), provision shall be made for hand cranking and a hand crank shall be furnished.

3.8.6 Charging system. Unless a QPL engine is furnished, the charging generator or alternator and voltage regulator shall be the supplier's standard. If an alternator is furnished, a positive device shall be

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provided to prevent damage to the alternator during emergency starting and battery charging. When specified (see 6.2), a waterproof, dual-socket receptacle conforming to MS75021 shall be provided to facilitate electrical connection between batteries and external power source for emergency charging or starting. Unless otherwise specified (see 6.2), a battery shall be furnished that is the supplier's standard for the size tractor supplied. The supplier's standard battery container shall be furnished and shall be mounted so that batteries contained therein will be level.

3.8.7 Exhaust cover. When a vertical exhaust is furnished, a counter balanced exhaust cover shall be furnished to prevent the entry of rain into the exhaust system.

3.8.8 Exhaust spark arrestor. When specified (see 6.2), the tractor shall be furnished with an exhaust spark arrestor to eliminate the emission of glowing sparks from the exhaust system.

3.9 Power train.

3.9.1 Transmission. The transmission shall be of the selective, sliding-gear, syncro-mesh or constant-mesh type. Transmission gears shall be of heattreated steel with machine-cut teeth. The input torque capacity of the transmission shall be greater than the maximum gross torque developed by the engine. When specified (see 6.2), a torque converter shall be furnished and the torque converter shall have a torque input rating of not less than 125 percent of the maximum torque output of the engine.

3.9.1.1 Speeds. Tractor sizes R-10 through R-30 shall have not less than three forward speeds. Tractor sizes R-35 through R-60 shall have not less than four forward speeds. All sizes of tractors shall have not less than one reverse speed. The size R-15 through R-60 tractors shall have the speeds shown in table I at maximum governed speed.

Table I. Speeds

Tractor size	Maximum speed (low gear) (mph)	Minimum speed (high gear) (mph)
R-15 through R-30	2	10
R-35 through R-60	2.5	12

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3.9.2 Clutch. When a flywheel clutch is required, it shall be the supplier's current standard type. The clutch shall withstand repeated starting and stopping without slipping, overheating, breakage, or malfunctioning when tested as specified in 4.7.6.

3.10 Drawbar horsepower. Each tractor shall develop drawbar horsepower falling within the size range specified in 1.2 when tested as specified in 4.7.2.

3.11 Towing. The tractor shall perform the following towing operations without malfunction, damage, or permanent deformation under all conditions specified herein:

(a) Tow a drag-type load requiring 90 percent of the maximum rated drawbar pull at the maximum speed in low gear as specified in table I, when tested as specified in 4.7.6.1.
(b) Tow a drag-type load requiring 35 percent of the maximum rated drawbar pull at the maximum speed in high gear as specified in table I, when tested as specified in 4.7.6.2.

3.12 Wheels and rims. Wheels shall conform to SAE J712. Rims shall be in accordance with the Tire and Rim Association Yearbook for the tire size required.

3.13 Tires. Tires shall be tube or tubeless with rib tread on the front tires and type R1, R3, or R4 tread as specified (see 6.2) on the rear tires. Commercial antiozonant shall be included in the tires during manufacture. Tires shall be of a rating in accordance with the Tire and Rim Association Yearbook to sustain any load imposed by the mounted attachments specified herein.

3.14 Rear fenders. The tractor shall be furnished with the supplier's current standard production rear fenders to prevent both splash and hurled object damage.

3.15 Brakes. Tractor rear wheels shall be provided with hydraulic or mechanical service brakes with or without vacuum booster. The service brakes shall be either simultaneously or independently controlled at the choice of the operator. The service brakes shall bring the tractor to a full stop within a distance of 15 feet from a speed of 12 mph on a smooth, hard, level, dry surface. Tractors having a maximum speed of less than 12 mph shall stop within the same number of feet distance as their maximum number of mph. A positive-locking, mechanical-type, parking brake shall be provided. The parking brake shall hold

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the tractor on a 30 percent longitudinal slope both upgrade and downgrade. Service brakes may be utilized as parking brakes if a positive lock or locks are provided. The brake-lever pull required to apply the parking brake shall not exceed 90 pounds.

3.16 Steering. The tractors shall be equipped with automotive-type steering on the front wheels. When specified (see 6.2), power-assist or full power steering shall be provided. The power-assist or full power steering mechanism shall preserve the features of directional stability inherent in the manually operated steering mechanism and shall incorporate manual steering which can be used for emergency when the power-assist or full power steering is inoperative. Precision adjustments to compensate for wear in all major steering parts shall be provided.

3.17 Hydraulic system. A hydraulic system shall be furnished on the tractor when any attachment is furnished which requires hydraulic control. The hydraulic system shall be the supplier's standard for the tractor and attachments except as otherwise specified herein. All components shall be located in a protected position to prevent damage from objects encountered in normal operation. Oil for the hydraulic system shall conform to MIL-L-2104, grade 10, or MIL-L-45199, grade 10. All component materials shall be compatible with these oils and the components shall operate as specified herein throughout an oil operating temperature range of minus 25° F. to plus 225° F. For low-temperature operation, oil conforming to MIL-L-10295 may be used. The temperature rise of the oil in any portion of the hydraulic system shall not exceed 100° F. above ambient temperature during all operations specified herein.

3.17.1 Cylinders. All hydraulic cylinder piston rods shall be plated in accordance with QQ-C-320, class 2. Plating shall be not less than 0.0005 inch thick.

3.17.2 Reservoir. The hydraulic oil reservoir shall have sufficient capacity to prevent a temperature rise of the oil above 100° F. ambient temperature during any operation specified herein. The reservoir shall be either pressurized or nonpressurized. If a nonpressurized reservoir is furnished, a breather filter shall be required. An oil level dipstick with "Add" and "Full" marks engraved or embossed on one side shall be furnished for checking the oil level in the reservoir. A magnetic drain plug and drain shall be furnished and located so as to allow draining of oil from the reservoir. A removable 40-mesh wire strainer shall be provided in the filler opening. The filler opening shall be of sufficient size to strain by gravity the oil specified herein at 70° F. at the rate of 2 gallons per minute.

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3.17.3 Hydraulic system oil filter. A full-flow oil filter shall be provided. The oil filter shall consist of a metal container and a replaceable element or shall be a one-piece disposable type constructed so that no special equipment or skill will be required in changing the filter elements. The filter shall have a nominal rating of 20 to 25 microns.

3.18 Accessories. Unless otherwise specified (see 6.2), the tractor shall be furnished with the following accessories.

3.18.1 Hitch. A quick-coupling suspension hitch. Unless otherwise specified (see 6.2), the hitch shall conform to SAE J715B.

3.18.2 Power takeoff. A rear power takeoff (PTO). The PTO shall conform to SAE J718a except that dimensional variations are acceptable, provided that the PTO will accept and be fully operable with all rear-mounted equipment conforming to applicable SAE standards. Unless otherwise specified (see 6.2), the PTO shall be the independent type conforming to SAE J722. A PTO drive conforming to SAE J721(b) shall be furnished.

3.18.3 Drawbar. A drawbar conforming to SAE J718a. Unless otherwise specified (see 6.2), the drawbar shall be of the fixed type.

3.19 Operator's compartment.

3.19.1 Seat. A seat shall be provided on the tractor for the operator. The seat shall be a shock-absorbing type which will effectively dampen the effect of road shock to the operator. The seat shall be adjustable for leg length of the operator. The seat and backrest shall be padded and covered with waterproof material. When specified (see 6.2), the supplier's standard seat shall be furnished.

3.19.2 Instrument panel. An instrument panel shall be provided. The panel shall be mounted so as to be visible from the operator's seat. An oil pressure gage, fuel gage, water temperature gage, light switches, starter control, ammeter, key operated ignition switch (class 1 only), hand throttle, and hand choke (class 1 only) shall be provided and mounted within reach of the operator. When specified (see 6.2), the oil pressure, ammeter, and temperature indicators shall be warning lights. When specified (see 6.2), a tachometer-hourmeter shall be furnished.

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3.19.2.1 Instrument light. An instrument light operated in conjunction with headlights shall be provided for the illumination of the instrument panel.

3.19.3 Trouble light. When specified (see 6.2), a trouble lamp conforming to Drawing B209297 with plug and weatherproof socket shall be provided. The socket shall be mounted in accordance with SAE standards in an accessible location on the instrument panel.

3.20 Lighting equipment. The tractors shall be equipped for night operation as specified herein. Grounded electrical circuits shall be used for the lighting equipment. Unless otherwise specified (see 6.2), cable and encasement shall be the supplier's standard. Terminal lugs shall be attached to the wire ends. Junction blocks shall have bases of thermosetting, laminated, phenol formaldehyde plastic or other equally suitable material and shall be equipped with studs, washers, and nuts for the attachment of the terminal lugs. The circuits shall be color or number traced. Grommets and clamps, to prevent chafing of the cable, shall be furnished where the wire passes through structural members. All wiring shall be supported as necessary to prevent chafing or other damage.

3.20.1 Headlights, taillight, and floodlight. Two headlights, one red taillight, and one adjustable rear floodlight shall be installed on each tractor.

3.20.1.1 Light housings. Unless otherwise specified (see 6.2), pedestal-mounted head and floodlight housings shall be in accordance with Drawing D13200E7450. Adjustable flush-type headlight housings with rubber sealed beam unit mountings are acceptable providing the mountings are in accordance with Drawing D13200E7450. Head and floodlights shall adjust and lock in both the horizontal and vertical azimuth. Taillights shall be weatherproof with vibration and shock resistant mountings.

3.21 Attachments. The tractors shall be furnished with loader, backhoe, and mower attachments, as specified (see 6.2). The attachments shall operate as specified herein. Self sealing, quick-disconnect couplings shall be furnished on all hydraulic line disconnect points.

3.21.1 Loader. The tractor shall be furnished with all necessary components for operation of the tractor with a loader. The loader attachment shall conform to table II. The loader bucket shall have a replaceable cutting edge. When loaders are furnished with tractors,

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the tractors shall have heavy-duty front axle and power steering conforming to 3.16. When specified (see 6.2), a grille guard shall be furnished with the loader. When a loader is the only attachment furnished, counter ballast as specified (see 6.2) shall be furnished. The operating controls shall be located so the operator can operate the bucket or other attachments with his right hand while steering with his left hand. Lift cylinders shall be double acting. The loader bucket shall be self leveling. When specified (see 6.2), the loader arms shall be compatible for use with other attachments such as a snowplow, a crane, or a pallet fork.

Table II. Loader Attachment Characteristics

Characteristic*	Tractor Size										
	R-10	R-15	R-20	R-25	R-30	R-35	R-40	R-45	R-50	R-55	R-60
(A) Minimum bucket dump clearance (in.)	--	75	75	80	98	98	100	100	102	102	102
(B) Minimum bucket reach (In.)	18	22	22	24	25	22	22	22	22	30	30
(C) Minimum digging depth below ground (In.)	1½	3	3	3	3	3	3½	4	4	5	5
(D) Minimum dumping angle (degrees)	--	45	45	45	45	45	45	45	45	45	45
(E) Minimum rollback at ground level (degrees)	14	15	15	15	14	15	15	15	20	20	20

*Characteristics are illustrated in figure 1.

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Table II. Loader Attachment Characteristics (Cont'd)

Characteristic*	Tractor Size										
	R-10	R-15	R-20	R-25	R-30	R-35	R-40	R-45	R-50	R-55	R-60
(F) Bucket hinge pin height	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(G) Minimum break-out force (lb)	600	600	800	1000	3000	3000	3000	4000	4000	5000	5000
(H) Minimum lift capacity (lb)	300	450	500	500	2000	2000	2000	3000	3000	3800	4000
(I) Bucket capacity heaped (cu ft) unless otherwise specified (see 6.2)	2½	5	7	7	10	10	15	15	15	18	18
(J) Maximum bucket dumping time (sec)	5	3	3	3	1.9	4	4	4	4	2.5	2.5
(K) Maximum bucket rollback time (sec)	-	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
(L) Maximum lifting time to full height (sec)	4	5	6½	6½	7	7	7	7	7	7	7
(M) Maximum lowering time (sec)	2	5	5	4.5	4.5	4.5	4.5	5	5	5	5

*Characteristics are illustrated in figure 1.

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Definitions of terms are as follows:

(A) Bucket dump clearance. Minimum clearance in inches, measured from ground level to bucket cutting edge, bucket dumped to 45 degrees.

(B) Reach. The distance measured from foremost point of tractor (including tires) to a vertical line from bucket cutting edge when dumping angle is 45 degrees at maximum height.

(C) Digging depth below ground level. Measured in inches bucket will dig below ground level with bucket bottom horizontal.

(D) Dumping angle. The angle that the bucket cutting edge will travel below horizontal when dumping at maximum height.

(E) Minimum rollback at ground. The vertical angle measured in degrees from cutting edge at maximum rollback position to the horizontal plane.

(F) Bucket hinge pin height. Distance from center of bucket hinge pin to ground line when bucket is at maximum height.

(G) Breakout force. Vertical upward force exerted at tip of cutting edge with the bucket rolling backward on the ground under the following conditions:

(1) Tractor on hard level surface with transmission in neutral.

(2) Relief valve at recommended setting.

(H) Lift capacity - full height. The maximum load (pounds) loader will lift to full height with load supported at center of gravity of bucket and positioned to retain maximum load. (Tractor to be stationary but not anchored.)

(I) Bucket capacity (cubic feet). SAE rating (nominal heaped).

(J) Bucket dumping time. Time required to rotate bucket from full rollback position to dump position.

(K) Bucket rollback time. Time required to rotate bucket from dump to rollback position.

(L) Lifting time to full height. Length of time for the lift arms to raise the empty bucket to full height from the normal loading position at ground level.

(M) Lowering time. Length of time for the lift arms to lower the empty bucket from full height to the normal loading position at ground level.

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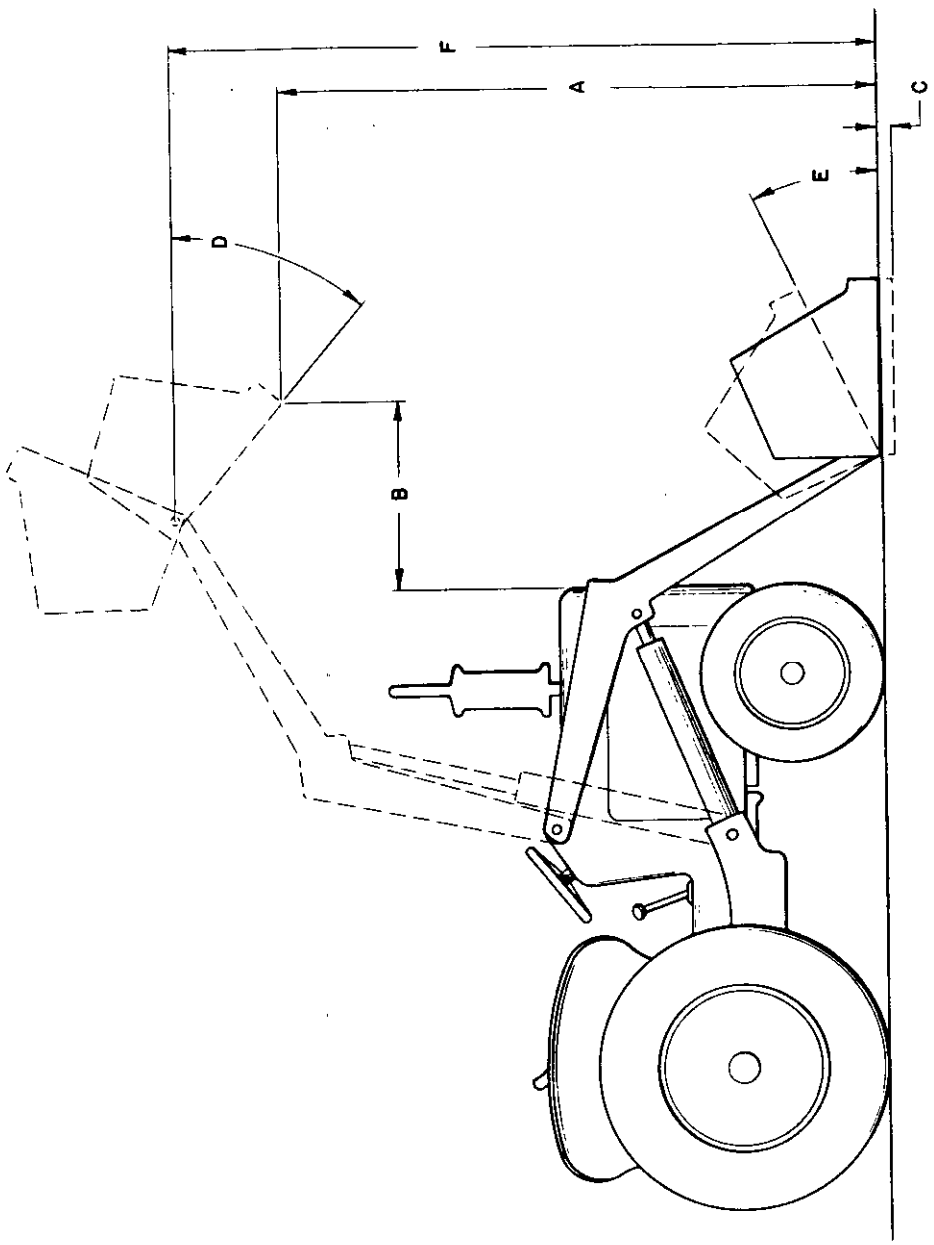


Figure 1. LOADER CHARACTERISTICS

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3.21.2 Backhoe. The tractor shall be furnished with all necessary components for operation of the tractor with a backhoe. The backhoe attachment shall conform to table III. When the backhoe is the only attachment supplied, counter ballast shall be furnished. The amount of counter ballast shall be as specified (see 6.2). The particular type of application will determine the width of the backhoe bucket. The backhoe shall be equipped with hydraulically-controlled stabilizers to permit straight and level holes on 10-degree slopes. When specified (see 6.2), the backhoe shall have not less than three available positions for offset work such as flush work against a building.

Table III. Backhoe Attachment Characteristics

Characteristics*	Tractor Sizes					
	R-30	R-35	R-40	R-45	R-50	R-55 and R-60
(A) Minimum digging depth (ft)	10	10	12	13	13-1/2	13-1/2
(B) Minimum reach (ft)	14	14	14	14	16	16
(C) Minimum loading height (ft-in.)	8-3	8-3	8-3	8-3	8-3	11
(D) Minimum swing (degrees)	180	180	180	180	180	180
(E) Minimum digging force (lb)	5500	5500	6000	8500	8500	8500
(F) Minimum bucket arc (degrees)	130°	130°	130°	130°	145°	145°
(G) Maximum swing time (sec)	4	4.5	4.5	3.5	3-1/2	3.5
(H) Minimum bucket rollback (degrees)	45	45	45	45	45	45
(I) Minimum bucket width (in.) unless otherwise specified (see 6.2).	12	12	12	18	18	18

*Characteristics are illustrated in figure 2.

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Definitions of terms are as follows:

(A) Digging depth. The vertical distance below ground level that backhoe will dig a flat bottom trench 2 feet in length. Measurement to be made at bucket edge excluding teeth.

(B) Reach. The horizontal distance at ground level from center of boom swing pivot to farthest point bucket cutting edge (excluding teeth) will reach.

(C) Loading height. Measured as maximum clearance under lowest point of bucket at maximum height with bucket fully closed.

(D) Swing. Measured in degrees of uninterrupted rotation about bucket hinge pin.

(E) Digging force. Maximum force applied at bucket cutting edge tangent to arc of bucket rotation. (Measurement to be taken without prying on heel of bucket and without consideration of weight of tractor and backhoe.)

(F) Bucket arc. The arc traversed by the bucket cutting edge when the bucket cylinder travels its full stroke.

(G) Swing time. The time for the boom to traverse through its full swing.

(H) Bucket rollback. The amount that the bucket opens with respect to the bucket arm. It is the outer section of the bucket arc measured between a line through the bucket hinge pin to the bucket blade base and a line through the bucket hinge pin to the bucket arm hinge pin.

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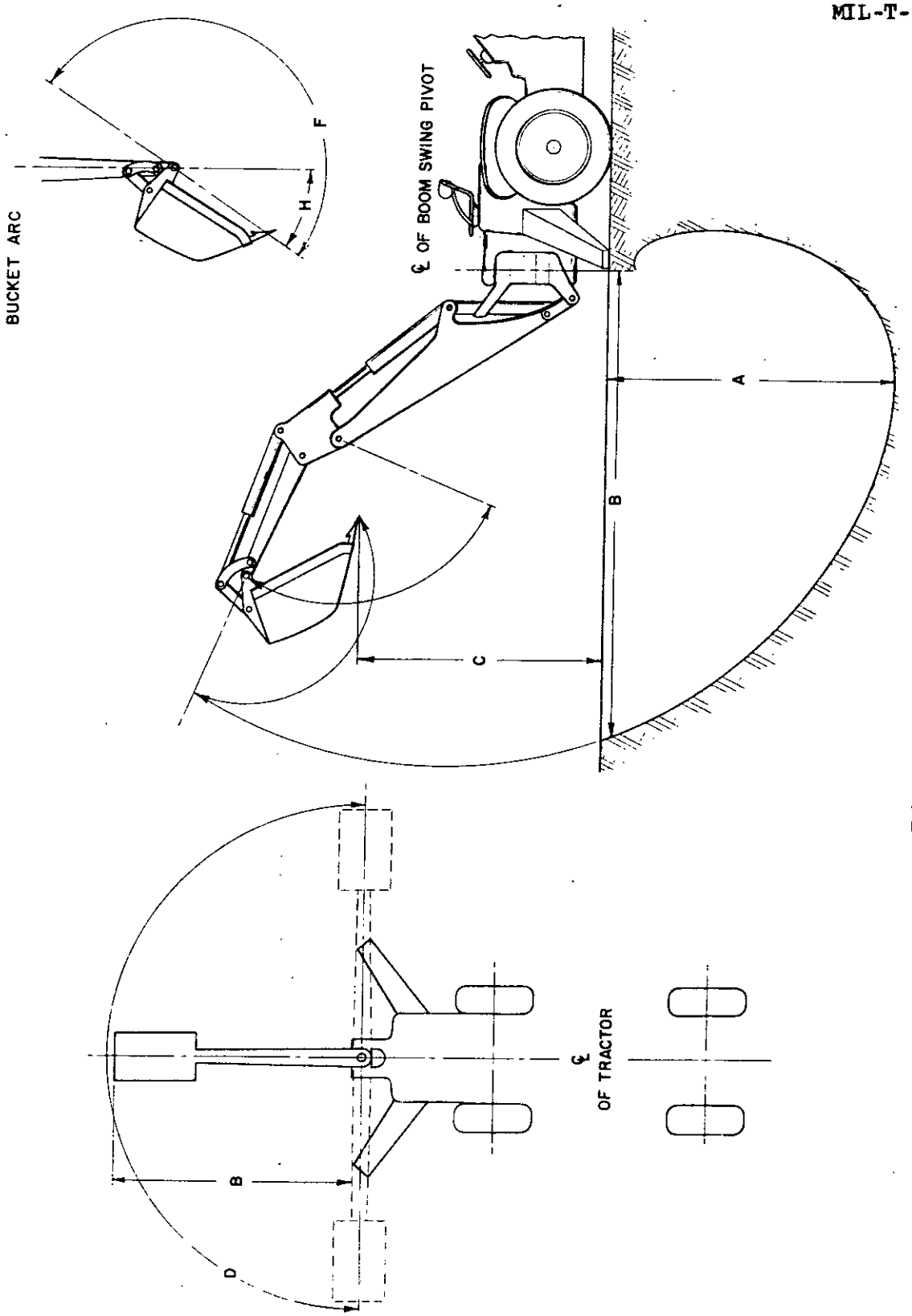


FIGURE 2. BACKHOE CHARACTERISTICS

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3.21.3 Mower. The tractor shall be furnished with all necessary components for operation of the tractor with the mowers specified herein.

3.21.3.1 Cutter bar mowers. Unless otherwise specified (see 6.2), cutter bar mowers shall be the side-mounted type. The cutter bar mowers shall be provided with a means to permit the selection of any cutter bar operating position by the driver while seated without stopping either the tractor or cutter bar. Provision shall be made to raise the inner shoe not less than 8 inches (curb lift) while mowing throughout its range of angularity. Means to tilt the cutter bar leading edge not less than 8 degrees below and above horizontal shall be provided. A heavy-duty break-back latch or similar release shall be provided to prevent damage to the cutter bar and sickle when driven against obstructions. A means for limiting the rearward travel of the cutter bar, to prevent the cutter bar from passing under the rear wheel, shall be provided. The cutter bar shall be equipped with inboard and outboard glides and sickle guards. Ledger plates, sickle blades, and other wearing parts shall be replaceable. The first cutter surface shall be just outside the tread of the rear tire. The drive shall have limited vibration as necessary to allow self cleaning. The drive mechanism shall be protected by a spring-loaded, dry-disk slip clutch. Unless otherwise specified (see 6.2), cutter bar mowers on size R-20 tractors and above shall have a width of cut of 7 feet and a cutting range from 75 degrees above to 45 degrees below horizontal.

3.21.3.2 Rotary mowers. Rotary mowers shall be the supplier's standard except as specified herein. Rotary mowers shall have metal hoods of not less than 3/16-inch thickness. The height of cut of the mower shall be adjustable from the operator's seat. The drive mechanism shall be protected by a spring-loaded, dry-disk slip clutch.

3.22 Radio-interference reduction. When specified (see 6.2), the tractors shall be equipped for the reduction of radio-interference in accordance with MIL-I-11683.

3.23 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.23.1 Lubricants. The tractor and attachments shall operate as specified herein when lubricated with Military lubricants.

3.23.2 Lubrication fittings. Lubrication fittings shall conform to MS15001, MS15002, or MS15003. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to

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MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.23.3 Pressure-release fitting. A pressure-release fitting shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.23.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the tractor is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground.

3.23.5 Lubrication service.

3.23.5.1 Grease lubrication. Bearing lubrication, including sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instrument and all sealed bearings and for other bearings where operating temperatures do not exceed 250° F.

(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F.

The tractor shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because Military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which Military grease has been used. The tag shall be a cloth tag conforming to UU-T-81, type A.

3.23.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), oil compartments shall be filled to the operating level with winter grades of Military oils designated for use in the temperature

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range of 0° F. to 32° F. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature ranges and grades of lubricants used. The tag shall be a cloth tag conforming to UU-T-81, type A.

3.24 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component or element surfaces previously cleaned and prepared.

3.25 Identification marking. The tractor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the tractor on plates conforming to MIL-P-514, type I, style 1, material conforming to GG-P-455, class 1.

3.25.1 Instruction plates. The tractor shall be equipped with instruction plates, including warnings and cautions suitably located, describing any special or important procedure to be followed in operating and servicing the tractor. Plates shall conform to MIL-P-514, type IX, material conforming to GG-P-455, class 1.

3.26 Treatment and painting. The portions of the tractors and their components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704.

3.27 Stenciling. The gross weight and center of gravity of the tractors shall be stenciled on each side of the tractor in such manner as to be discernible. The maximum and minimum allowable tire pressure as recommended by the Tire and Rim Association shall be stenciled on the fenders for the rear tires and on the side of the tractor for the

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front tires with block- or stencil-type letters not more than 1 inch and not less than 3/4 inch high. Unless otherwise specified herein, stenciling shall conform to MIL-T-704.

3.28 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

3.29 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.30 Toolbox. Unless otherwise specified herein, the tractor shall be furnished with a toolbox made of mild steel not less than 0.06 inch in nominal thickness. The toolbox shall have a hinged lid and a positive clamping device that will keep the cover closed when subjected to vibration. The toolbox shall be securely fastened to the tractor in a protected position. The toolbox shall be of a size to contain the tools specified in 3.29. The lid shall open at least 90 degrees from the closed position and shall remain in the fully open position by the weight of the lid or by a holding device. Provisions shall be made to drain the toolbox in its mounted position. When specified (see 6.2), the supplier's standard toolbox shall be furnished.

3.31 Items to be furnished with tractor. No shipment of tractors shall be made unless repair parts, maintenance tools, operating instructions, and accessories are included with the shipment or unless approval for shipment of the tractors without such items has been received from the contracting officer.

3.32 Workmanship. All parts, components, and assemblies of the tractor including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.8).
(e) Inspection of preparation for delivery (see 4.9).

4.3 Preproduction inspection. Prior to examination and test:

(a) The tractor shall be lubricated with the oils and greases specified in 3.23. Oils shall be those designated for use in the ambient temperature at which the test is being performed.
(b) The tractor shall be run-in as prescribed by the supplier.

4.3.1 Examination. Prior to being tested, the tractor shall be examined for the defects marked "X" in column 1 of table IV. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the tractor shall be subjected to the tests marked "X" in column 1 of table V. Failure of any test shall be cause for rejection.

4.4 Initial production inspection.

4.4.1 Examination. The initial production tractor shall be examined for the defects marked "X" in column 2 of table IV.

4.4.2 Tests. Upon successful completion of the examination specified in 4.4.1, the tractor shall be subjected to the tests marked "X" in column 2 of table V.

4.4.3 Inspection failure. Failure of the initial production tractor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4.1 and 4.4.2, respectively, shall

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be cause for refusal by the Government to accept production tractors until evidence has been provided by the supplier that corrective action has been taken to eliminate deficiencies.

4.5 Quality conformance inspection.

4.5.1 Individual examination. After successful completion of all tests specified in 4.5.2 each tractor shall be examined for the defects marked "X" in column 3 of table IV. Presence of one or more defects shall be cause for rejection.

4.5.2 Individual tests. Each tractor shall be subjected to the tests marked "X" in column 3 of table V. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table IV.

Table IV. Examination Schedule

Preproduction Pretest	Production		Defect		Requirements Paragraph
	Initial Production	Individual			
1	2	3	4		5
-	X	X	101.	Material not of same quality as used on preproduction model.	3
X	X	-	102.	All accessories supplied commercially as standard equipment not furnished.	3.1
X	X	X	103.	Materials not as specified.	3.4
X	X	X	104.	Threaded parts not in accordance with Handbook H-28.	3.4.1
X	-	-	105.	Environmental and use conditions not as specified.	3.5

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Table IV. Examination Schedule (Cont'd)

Preproduction Pretest	Production Initial Production	Production Individual	Defect	Requirements Paragraph
1	2	3	4	5
X	X	X	106. Exposed parts which are subject to high operating temperatures or which are energized electrically not insulated, enclosed, or guarded.	3.6
X	X	-	107. Moving parts of such nature or so located as to be a hazard to operating personnel not enclosed or guarded.	3.6
X	X	-	108. Protective devices impair operating function.	3.6
X	X	-	109. Ease of maintenance not as specified.	3.7
X	X	-	110. Gasoline engine not as specified.	3.8.1
X	X	-	111. Ignition system not as specified.	3.8.1.1
X	X	-	112. Diesel engine not as specified.	3.8.2
X	X	-	113. Power and speed ratings not as specified.	3.8.3
X	X	-	114. Fuel system not as specified.	3.8.4
X	X	-	115. Cranking system not as specified.	3.8.5

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Table IV. Examination Schedule (Cont'd)

Preproduction Pretest	Production Initial Production	Production Individual	Defect	Requirements Paragraph
1	2	3	4	5
X	X	-	116. Charging system not as specified.	3.8.6
X	X	X	117. Exhaust cover not furnished.	3.8.7
X	X	X	118. Exhaust spark arrestor not furnished.	3.8.8
X	X	-	119. Transmission not as specified.	3.9.1
X	X	-	120. Number of speeds for transmission not as specified.	3.9.1.1
X	X	-	121. Clutch not as specified.	3.9.2
X	X	-	122. Wheels and rims not as specified.	3.12
X	X	-	123. Tires not as specified.	3.13
X	X	-	124. Rear fenders not as specified.	3.14
X	X	-	125. Brakes not as specified.	3.15
X	X	-	126. Steering not as specified.	3.16
X	X	-	127. Hydraulic system not as specified.	3.17
X	X	-	128. Hydraulic cylinders not as specified.	3.17.1
X	X	-	129. Hydraulic reservoir not as specified.	3.17.2
X	X	-	130. Hydraulic system oil filter not as specified.	3.17.3

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Table IV. Examination Schedule (Cont'd)

Preproduction Pretest	Production Initial Production	Production Individual	Defect		Requirements Paragraph
1	2	3	4		5
X	X	-	131.	Hitch not as specified.	3.18.1
X	X	-	132.	Power takeoff not as specified.	3.18.2
X	X	-	133.	Drawbar not as specified.	3.18.3
X	X	-	134.	Seat not as specified.	3.19.1
X	X	-	135.	Instrument panel not as specified.	3.19.2
X	X	-	136.	Instrument light not as specified.	3.19.2.1
X	X	-	137.	Trouble light not as specified.	3.19.3
X	X	-	138.	Lighting equipment not as specified.	3.20
X	X	-	139.	Headlights, taillight, and floodlight not as specified.	3.20.1
X	X	-	140.	Light housings not as specified.	3.20.1.1
X	X	X	141.	Quick-disconnect couplings not furnished on all hydraulic line disconnect points.	3.21
X	X	-	142.	Loader bucket cutting edge not replaceable.	3.21.1
X	X	X	143.	Tractors equipped with loaders do not have heavy-duty front axle and power steering.	3.21.1

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Table IV. Examination Schedule (Cont'd)

Preproduction Pretest	Production Initial Production	Production Individual	Defect	Requirements Paragraph
1	2	3	4	5
X	X	-	144. Grille guard not furnished with loader when specified.	3.21.1
X	X	-	145. Operating controls not located so operator can control the bucket or other attachments with his right hand while steering with his left hand.	3.21.1
X	-	-	146. Loader bucket not capable of self leveling.	3.21.1
X	X	-	147. Stabilizers not as specified.	3.21.2
X	X	-	148. Backhoe does not have three available positions for offset work when specified.	3.21.2
X	X	-	149. Cutter bar mower not as specified.	3.21.3.1
X	X	-	150. Rotary mowers not as specified.	3.21.3.2
-	X	X	151. Radio interference reduction not as specified.	3.22
X	X	X	152. Lubrication not as specified.	3.23
X	X	-	153. Fungus and moisture resistance not as specified.	3.24
X	X	X	154. Identification marking not as specified.	3.25

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Table IV. Examination Schedule (Cont'd)

Preproduction Pretest	Production Initial Production	Production Individual	Defect	Requirements Paragraph
1	2	3	4	5
X	X	X	155. Treatment and painting not as specified.	3.26
X	X	X	156. Stenciling not as specified.	3.27
X	X	X	157. Technical publications not as specified.	3.28
X	X	-	158. Repair parts and maintenance tools not as specified.	3.29
X	X	-	159. Toolbox not as specified.	3.30
X	X	X	160. Items as specified not furnished with tractor.	3.31
X	X	X	161. Workmanship not as specified.	3.32

4.6.2 Tests.

4.6.2.1 Test sequence. All tests shall be conducted in accordance with table V in any sequence except the towing test (see 4.7.6) shall be next to the last and the post-test disassembly and examination (see 4.7.10) shall be last.

4.6.2.2 Test conditions. All test apparatus shall be accurate within the limits specified in MIL-STD-448, section 6, except as specified herein. Prior to test, the tractor shall be lubricated with oils and greases specified in 3.23. Oils shall be those designated for use in the ambient temperature at the place of test.

4.6.2.3 Test schedule. Tests shall be in accordance with table V.

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Table V. Test Schedule

Preproduction	Initial	Individual	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	Test conditions	4.6.2.2	-
-	-	X	Operational (tractor)	4.7.1	Section 3
X	X	-	Drawbar horsepower	4.7.2	3.10
X	X	-	Speed	4.7.3	3.9.1.1
X	X	-	Service brakes	4.7.4.1	3.15
X	X	-	Parking brake	4.7.4.2	3.15
X	-	-	Slope	4.7.5	3.5.2
X	-	-	Towing	4.7.6	3.11, 3.9.2
X	X	-	Radio-interference reduction	4.7.7	3.22
X	-	-	Environmental	4.7.8	3.5.1
-	-	X	Operational test for attachments	4.7.9.1	3.5.2
X	X	-	Loader	4.7.9.2	3.21.1
X	X	-	Backhoe	4.7.9.3	3.21.2
X	X	-	Cutter-bar mower	4.7.9.4	3.21.3.1
X	X	-	Rotary mower	4.7.9.5	3.21.3.2
X	-	-	Post-test disassembly and examination	4.7.10	Section 3

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4.7 Test procedure.

4.7.1 Operational (tractor). The tractor shall be run at full governed speed for not less than 1 hour to determine the functioning of such components as steering, transmission, clutch, and brakes. During the 1-hour operation, the transmission shall be shifted through its entire gear range at least 10 times. Ten emergency stops shall be made to determine the operating performance of the brakes. Evidence of any malfunction, inability of the tractor to run at full governed speed for 1 hour, or inability of any control to operate as specified herein shall constitute failure of this test.

4.7.2 Drawbar horsepower. The tractor shall be tested for drawbar horsepower in accordance with SAE J782. No ballast shall be allowed. Nonconformance to 3.10 shall constitute failure of this test.

4.7.3 Speed. With the engine running at full governed speed, the tractor shall be driven in low gear on a hard, level test course of not less than 200 feet in length. The speed traversing the course shall be determined. With the engine running at full governed speed, the tractor shall be driven in high gear on a hard, level test course for a distance of 600 feet. The speed traversing the course shall be determined. Failure of the tractor to traverse the courses at the speeds as specified in 3.9.1.1 shall constitute failure of this test.

4.7.4 Brake.

4.7.4.1 Service brakes. The test course shall be a dry, level, improved surface allowing no tire penetration and having a grade of not more than 1 percent. Attach a two-barrel detonator apparatus of the chalk- or paint-marking type to the tractor brake system. One barrel shall be actuated by a switch directly connected to the brake pedal and the second barrel shall be actuated by the driver or observer. When the tractor has attained a speed of 12 mph, apply the brakes, actuating the first barrel. After the tractor has come to a full stop, actuate the second barrel. Measure the distance between the point of brake application and the point of stop. Perform the test four times, two times in each direction. Nonconformance to 3.15 shall constitute failure of this test.

4.7.4.2 Parking brake. The test course shall be a hard, dry, smooth surface having a slope of not less than 30 percent and not less than 30 feet long. Position the tractor facing downgrade, apply the parking brake, and place the transmission in neutral. Repeat the test with the tractor facing upgrade. Nonconformance to 3.15 shall constitute failure of this test.

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4.7.5 Slope. Test the tractor as follows:

(a) Run the engine for 5 minutes at the supplier's recommended low idle with the crankcase oil level at the low mark on the oil-level indicator and with the right side of the tractor downhill on a 30 percent slope or simulated slope.

(b) Repeat the test with the left side of the tractor downhill.

(c) Repeat the test with the front of the tractor downhill on a 30 percent slope or simulated slope.

(d) Repeat the test with the rear of the tractor downhill on a 30 percent slope.

(e) Run the engine for 15 minutes at top governed speed with the crankcase filled to the full level and with the right side of the tractor downhill on a 30 percent slope or simulated slope.

(f) Repeat the test in (e) with the left side of the tractor downhill.

(g) Repeat the test in (e) with the front of the tractor downhill on a 30 percent slope or simulated slope.

(h) Repeat the test in (e) with the rear of the tractor downhill on a 30 percent slope or simulated slope.

Nonconformance to 3.5.2 shall constitute failure of this test.

4.7.6 Towing.

4.7.6.1 Low speed towing. Tow a drag-type load, requiring 90 percent of the supplier's maximum rated drawbar pull. Operate the tractor at maximum obtainable speed in low gear. Tow the load for a distance of 250 feet and stop the tractor for a 10-second period. Repeat the cycle every 250 feet for 5 continuous hours stopping only for necessary refueling. Clutch slippage or overheating, or breakage or malfunctioning of any part or component of the tractor shall constitute failure of this test.

4.7.6.2 High speed towing. Operate the tractor to tow a drag-type load requiring 35 percent of the supplier's maximum rated drawbar pull, at maximum obtainable speed. Tow the load 500 feet and stop the tractor

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for a 10-second period. Repeat this cycle every 500 feet for 5 continuous hours stopping only for necessary refueling. Operate the tractor in the highest gear possible. Clutch slippage or overheating, or breakage or malfunctioning of any part or component of the tractor shall constitute failure of this test.

4.7.7 Radio-interference reduction. The supplier is responsible for conducting all tests to determine conformance to MIL-I-11683. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer a report detailing the results of the tests. The test report shall include a block diagram of the test setup, a description of the interference reduction system applied, and a list of the test equipment utilized. The interference reduction system applied to the approved model shall be incorporated on all subsequent production tractors.

4.7.8 Environmental.

4.7.8.1 Low temperature. Subject the tractor, serviced with arctic lubricants, to a temperature of minus 25° F. until stabilized. Start the engine by cranking with the batteries which are on the tractor and which have been stabilized. Do not use substitute batteries or other external cranking energy. Fifteen minutes after starting, operate the engine under rated-load conditions. The operator shall wear arctic clothing, including mittens and boots, during this test. Inability to start the engine within 5 minutes, inability to operate under rated-load conditions within 15 minutes without unnatural or continued control manipulation, failure of controls and instruments to operate as specified after 15 minutes, or inability of the operator to manipulate any control or adjustable device because of the bulk of arctic clothing shall constitute failure of this test.

4.7.8.2 High temperature. Repeat the test specified in 4.7.8.1 at a temperature of plus 115° F., except as follows: Service the tractor with Military lubricants suitable for a temperature of plus 115° F. and with the operator dressed in normal clothing operate the engine under rated-load conditions for a period of 2 hours. Inability to start the engine within 5 minutes, inability to operate under rated-load conditions within 15 minutes without unnatural or continued control manipulation, inability to maintain engine temperatures within the ranges specified in MIL-E-11276, MIL-E-11275 or as recommended by the supplier for his standard engines, or inability of controls and instruments to operate as specified after 15 minutes shall constitute failure of this test.

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4.7.9 Attachments.

4.7.9.1 Operational test for attachments. Each tractor attachment shall be operated to determine the functioning of all components. If the attachment is to be delivered on the tractor, the attachment shall be mounted on the tractor and all controls of the attachment shall be operated throughout the entire range of movements. If the attachment is not delivered with a tractor, the attachment shall be assembled as a unit and bench tested to determine the adequacy of the parts and controls. Malfunctioning of any part or inability of any control to operate as specified herein shall constitute failure of this test.

4.7.9.2 Loader. Operate the tractor with loader attachment as follows:

(a) Load 75 heaped loads (from stockpiles) in a dump truck having a sideboard height equal to the minimum bucket dump clearance as specified in table II. A simulated truck sideboard may be used. During the loading, the loading characteristics as specified in table II shall be measured.
(b) Operate for 50 cycles moving a stockpile. A cycle shall consist of loading from a stockpile, carrying the load in forward gear not less than a 100-foot distance, dumping, and returning empty to the stockpile.

Inability of the tractor with loader attachment to operate as specified in (a) and (b) above, nonconformance to 3.21.1, evidence of breakage, or permanent deformation, or failure of any component or part shall constitute failure of this test.

4.7.9.3 Backhoe. The tractor, with backhoe attachment, shall be operated in undisturbed soil for not less than 2 hours digging to depths as specified in table III. Nonconformance to 3.21.2, evidence of breakage or permanent deformation, or failure of any component or part shall constitute failure of this test.

4.7.9.4 Cutter bar mower. The tractor, with cutter bar mower attachment, shall be operated for not less than 2 hours cutting grass and weeds. Nonconformance to 3.21.3.1, evidence of breakage or permanent deformation, or failure of any component or part shall constitute failure of this test.

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4.7.9.5 Rotary mower. The tractor with rotary mower attachment shall be operated for not less than 2 hours cutting heavy grass, thick weeds, and brush. Evidence of malfunctioning, permanent deformation or breakage, or failure of any component or part shall constitute failure of this test.

4.7.10 Post-test disassembly and examination. When specified (see 6.2) and upon completion of all tests, the tractor shall be dismantled and the following components shall be disassembled in detail and examined:

(a) Engine - The cylinder head, all valves, two pistons and piston pins removed from the two cylinders having the lowest compression pressure reading at the end of the test, two main bearings, and two connecting rod bearings removed.
(b) Transmission.
(c) Torque converter, when applicable.
(d) Differential.
(e) Drive shafts and U-joints.
(f) Brake system (complete).
(g) Power steering system.

(1) All cylinders
(2) All valves
(3) Pump

(h) Hydraulic system.

(1) Pumps and valves
(2) All cylinders
(3) Controls
(4) Hose and fittings where leakage has occurred.

(i) Wheels and bearings.

Additional disassembly and examination shall be made at the option of the contracting officer. A complete description of the condition of all components and parts of the tractor shall be recorded, and photographs shall be taken of any deficiency. Each evidence of permanent deformation, breakage, or excessive wear of any part or component shall constitute a defect. The contracting officer shall have the final determination as to what constitutes excessive wear.

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4.8 Inspection comparison. Unless otherwise specified (see 6.2) the Government may select tractors at any time during the contract production period and subject the tractors to the examination specified in columns 1 and 3 of table IV and to the tests specified in column 2 of table V. The inspection shall be performed by the Government at a site selected by the Government. Tractors will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model or initial production tractor.

4.8.1 Inspection failure. Failure of any tractor when examined and tested as specified in 4.8 may be considered cause for refusal by the Government to continue acceptance of tractors until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.9 Inspection of preparation for delivery.

4.9.1 Quality conformance inspection of pack.

4.9.1.1 Unit of product. For the purpose of inspection, a completely processed tractor prepared for shipment shall be considered a unit of product.

4.9.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.9.1.3 Examination. Samples selected in accordance with 4.9.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

162.	Materials, methods, and containers not as specified for level A and B. Each incorrect material, method, or container shall be considered one defect.
163.	Preservation and packaging not in accordance with the referenced document for level A.
164.	Preservation and packaging not as specified for level B.
165.	Packing not in accordance with the referenced document for level A and B.
166.	Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

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5.1.1 Level A. The tractor, attachments, components, repair parts, tools, and publications shall be preserved and packaged in accordance with the level A requirements of MIL-T-3351.

5.1.2 Level B. The tractor, attachments, components, repair parts, tools, and publications shall be preserved and packaged in accordance with the level A requirements of MIL-T-3351, with the following exceptions.

5.1.2.1 Identification and instruction plates. Unpainted metal identification and instruction plates shall not require coating with varnish.

5.1.2.2 Engine. The engine fuel system, combustion chambers, valves, and liquid cooling system only shall be preserved in accordance with the level A requirements of MIL-E-10062, type III.

5.1.2.3 Clutches and brakes. Coating of clutch and brake components with primer shall not be required.

5.1.2.4 Lamp assemblies and trouble light. Removed lamp assemblies and the trouble light shall be preserved together in accordance with MIL-P-116, method III, using a box conforming to PPP-B-636, type CF, class domestic.

5.1.3 Level C. The tractor, attachments, components, repair parts, tools, and publications shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination in accordance with the supplier's recommendation subject to the approval of the contracting officer (see 6.3).

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The tractor, attachments, components, repair parts, tools, and publications shall be packed in accordance with the level A requirements of MIL-T-3351, type I, II, or III as specified (see 6.2).

5.2.2 Level B. The tractor, attachments, components, repair parts, tools, and publications shall be packed in accordance with the level B requirement of MIL-T-3351, type I or III as specified (see 6.2).

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5.2.3 Level C. The tractor, attachments, components, repair parts, tools, and publications shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates. Containers shall comply with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 Marking. Packages, shipping containers, and unboxed (mobile) tractors shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tractors covered by this specification are intended for use with attachments such as front-end loaders, backhoes, and cutter bar mower, and for drawbar work and other industrial use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size required (see 1.2).
(c) When a preproduction model is not required (see 3.2).
(d) When an initial production tractor is required (see 3.3).
(e) When a gasoline engine conforming to MIL-E-11275 is required (see 3.8.1).
(f) When diesel engine shall conform to MIL-E-11276 (see 3.8.2).
(g) When provision for hand cranking is required (see 3.8.5).
(h) When a waterproof dual-socket receptacle conforming to MS75021 is required (see 3.8.6).
(i) When the battery will be other than the supplier's standard (see 3.8.6).
(j) When an exhaust spark arrestor is required (see 3.8.8).
(k) When a torque converter is required (see 3.9.1).
(l) Type tread required on the rear wheels (see 3.13).
(m) When power assist or full power steering is required (see 3.16).
(n) When and what accessories are not required (see 3.18).
(o) When the hitch shall not conform to SAE J715B (see 3.18.1).
(p) When other than an independent pto is required (see 3.18.2).
(q) When other than the fixed type drawbar is required (see 3.18.3).
(r) When supplier's standard seat shall be furnished (see 3.19.1).

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(s) When oil pressure, ammeter, and temperature indicators shall be warning lights (see 3.19.2).
(t) When a tachometer-hourmeter is required (see 3.19.2).
(u) When a trouble lamp is required (see 3.19.3).
(v) When cable and encasement shall be other than the supplier's standard (see 3.20).
(w) When light housings shall not be in accordance with Drawing D13200E7450 (see 3.20.1.1).
(x) Attachments required (see 3.21).
(y) When a grille guard is required (see 3.21.1).
(z) Counter ballast required with loader (see 3.21.1).

(aa) When loader arms shall be compatible for use with other attachments (see 3.21.1).
(bb) When bucket capacity shall be other than that specified in table II.
(cc) Ballast required with backhoe (see 3.21.2).
(dd) When backhoes shall have at least three available positions (see 3.21.2).
(ee) When backhoe bucket widths shall be other than that specified in table III.
(ff) When cutter bar mower shall be other than the side-mounted type (see 3.21.3.1).
(gg) When other than a 7-foot cutter width is required (see 3.21.3.1).
(hh) When radio interference suppression is required (see 3.22).
(ii) When fungus and moisture resistance are not required (see 3.24).
(jj) Technical publications required (see 3.28).
(kk) Repair parts and maintenance tools required (see 3.29).
(ll) When the supplier's standard toolbox is required (see 3.30).
(mm) When post-test disassembly and examination is required (see 4.7.10).
(nn) When inspection comparison test will not be conducted (see 4.8).
(oo) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the tractors will be held in covered storage for an indefinite period.
(pp) When type I, II, or III packing in accordance with MIL-T-3351 is required (see 5.2.1 and 5.2.2).

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6.3 Prior to start of production preservation and packaging, the supplier should be required to submit his preservation and packaging proposals, in detail, to the contracting officer for approval (see 5.1.3).

6.4 Radio-interference reduction tests. The contracting officer should submit a copy of the report required by 4.7.7 to the Director, U. S. Army Electronics Laboratories, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey.

6.5 Classification designation changes. The following list indicates the relationship between the sizes specified in MIL-T-3229A and in this specification:

This specification	MIL-T-3229A
Size R-10	None
Size R-15	None
Size R-20	Size 4
Size R-25	Size 4
Size R-30	Size 5
Size R-35	Size 5
Size R-40	Size 5
Size R-45	None
Size R-50	None
Size R-55	None
Size R-60	None

Custodians:

Army - MO(ERDL)
Navy - YD
Air Force - 84

Review activities:

Army - MO
Navy - None
Air Force - 84, 85

User activities:

Army - None
Navy - YD, MC
Air Force - None

Preparing activity:

Army - MO(ERDL)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-T-3229B; Tractors, Wheeled, Industrial, Pneumatic-Tired, Gasoline- and Diesel-Engine-Driven

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *IF "YES", IN WHAT WAY?*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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